

Neeloy Chakraborty

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Research Statement:

As a fifth year PhD candidate at UIUC, my research goal is to develop reliable technologies to enable safer interactions between humans and automation. To do so, I have:

- 1. studied modular approaches for **instruction-following embodied AI** robots*
- 2. developed **real-time video generation models** to allow users to remotely control robots under challenging constraints*
- 3. designed scalable algorithms to **detect anomalies in multi-agent settings***
- 4. identified **hallucinations in large language model** generations to proactively avoid failures in decision-making*

Interests: Embodied AI, Generative AI, Robot Teleoperation, Anomaly and Hallucination Detection, Reinforcement Learning

Skills: Python, PyTorch, PyTorch3D, Ollama, OpenCV, Blender, ROS, ZED Camera, Linux, Git, CARLA, Docker, C/C++

Education:

- University of Illinois Urbana-Champaign**
 - PhD in Electrical and Computer Engineering (Exp. May 2026) GPA 3.90/4.00
 - Master of Science in Electrical and Computer Engineering (2021 – 2023) [Thesis] GPA 3.83/4.00
 - Bachelor of Science in Computer Engineering (2017 – 2021) [Thesis] GPA 3.75/4.00

Current Research Projects:

- Creative 3D Asset Generation** Summer 2025 –
Advisor: Dr. Guan-Ming Su
Developing 3D asset generation models conditioned on different modalities
Dolby Laboratories
- Hallucination Detection and Mitigation for Foundation Models in Decision-Making Tasks** Spring 2024 –
Advisors: Professor Melkior Ornik and Professor Katherine Driggs-Campbell
Formulating pipelines to identify hallucinations in large (visual) language models
University of Illinois

Past Research Projects:

- Image Generation for Delay-Compensated Video Feeds in Mobile Robot Teleoperation** Summer 2023 – Fall 2024
Advisor: Professor Katherine Driggs-Campbell
Developing real-time image generation methods combining depth foundation models and neural rendering
University of Illinois
- Co-operative Traffic Congestion Mitigation** Spring 2022 – Spring 2024
Advisors: Professor Cathy Wu and Professor Katherine Driggs-Campbell
Evaluating human-in-the-loop traffic congestion mitigation policies with real users in a car simulator
University of Illinois
- Highway Vehicle Anomaly Detection** Fall 2021 – Spring 2024
Advisor: Professor Katherine Driggs-Campbell
Designing frameworks for identifying anomalies on roads from multi-modal sensor data
University of Illinois
- Modular Embodied Agent for Alexa Arena** Fall 2022 – Spring 2023
Advisor: Professor Julia Hockenmaier
Leading development of an instruction-following embodied agent with action prediction and computer vision
University of Illinois

- **Decentralized Robot Crowd Navigation** Fall 2019 – Spring 2023
Advisor: Professor Katherine Driggs-Campbell University of Illinois
Researching a novel network to guide a robot to reach a goal state while avoiding colliding with other agents
 - **Koopman Models for Reinforcement Learning** Summer 2022
Advisors: Dr. Kaushik Balakrishnan and Dr. Devesh Upadhyay Ford Motor Company
Exploring benefits of combining Koopman theory with model-based RL for driving tasks
 - **Hierarchical Self-Imitation Reinforcement Learning with Sparse Rewards** Fall 2020 – Spring 2021
Advisor: Professor Katherine Driggs-Campbell University of Illinois
Applying hierarchical and self-imitation learning to long horizon single-agent environments with sparse rewards
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Publications

(* and + denote equal contribution and noteworthy comments are in red)

Pre-prints:

- **N. Chakraborty**, J. Pohovey, M. Ornik, and K. Driggs-Campbell. “Adaptive Stress Testing Black-Box LLM Planners,” Under review 2025 [[Paper](#)]

Journal Articles:

- S. Liu, H. Xia, F. C. Pouria, K. Hong, **N. Chakraborty**, Z. Hu, J. Biswas, and K. Driggs-Campbell. “HEIGHT: Heterogeneous Interaction Graph Transformer for Robot Navigation in Crowded and Constrained Environments,” IEEE Transactions on Automation Science and Engineering (T-ASE), 2025 [[Paper](#)] [[Website](#)] [[Code](#)]
- A. Hasan, **N. Chakraborty**, H. Chen, C. Wu, and K. Driggs-Campbell. “Lessons in Cooperation: Driver Sentiments towards Real-Time Advisory Systems,” IEEE Intelligent Transportation Systems Magazine (ITSM), 2025 [[Paper](#)]
- **N. Chakraborty**, M. Ornik, and K. Driggs-Campbell. “Hallucination Detection in Foundation Models for Decision-Making: A Flexible Definition and Review of the State of the Art,” ACM Computing Surveys (CSUR), 2025 [[Paper](#)]
- A. Hasan, **N. Chakraborty**, H. Chen, J-H. Cho, C. Wu, and K. Driggs-Campbell. “Cooperative Advisory Residual Policies for Congestion Mitigation,” ACM Journal on Autonomous Transportation Systems (JATS), 2024 [[Paper](#)]
- T. Ji*, **N. Chakraborty***, A. Schreiber, and K. Driggs-Campbell. “An Expert Ensemble for Detecting Anomalous Scenes, Interactions, and Behaviors in Autonomous Driving,” International Journal of Robotics Research (IJRR), 2024 [[Paper](#)]

Conference Articles:

- **N. Chakraborty***, Y. Fang*, A. Schreiber, T. Ji, Z. Huang, A. Mihigo, C. Wall, A. Almana, and K. Driggs-Campbell. “Towards Real-Time Generation of Delay-Compensated Video Feeds for Outdoor Mobile Robot Teleoperation,” IEEE International Conference on Robotics and Automation (ICRA) 2025 [[Paper](#)] [[Website](#)] [[Code](#)] [[Dataset](#)] (Leading cross-departmental team of researchers with diverse skillsets)
- P. Chang, S. Liu, T. Ji, **N. Chakraborty**, K. Hong, and K. Driggs-Campbell. “A Data-Efficient Visual-Audio Representation with Intuitive Fine-tuning for Voice-Controlled Robots,” Conference on Robot Learning (CoRL) 2023 [[Paper](#)] [[Website](#)]
- A. Hasan, **N. Chakraborty***, H. Chen*, J-H. Cho, C. Wu, and K. Driggs-Campbell. “PeRP: Personalized Residual Policies for Congestion Mitigation Through Co-operative Advisory Systems,” IEEE International Conference on Intelligent Transportation Systems (ITSC) 2023 [[Paper](#)] [[Website](#)] [[Code](#)]
- **N. Chakraborty***, R. Sidhu*, B. Abdullahi*, H. Chen*, N. Ravi*, A. Ankur, D. Prasad, and J. Hockenmaier. “BEAST: Building an Embodied Action-prediction System with Trajectory data,” Alexa Prize SimBot Challenge Proceedings 2023 [[Paper](#)] [[Website](#)] (University team leader and semi-finalist in inaugural Amazon SimBot Challenge)

- S. Liu, P. Chang, Z. Huang, **N. Chakraborty**, K. Hong, W. Liang, D. L. McPherson, J. Geng, and K. Driggs-Campbell. "Intention Aware Robot Crowd Navigation with Attention-Based Interaction Graph," IEEE International Conference on Robotics and Automation (ICRA) 2023
[Paper] [Website] [Code] **(Best poster award at IROS 2023 Last-Mile Robotics Workshop)**
- **N. Chakraborty**, A. Hasan*, S. Liu*, T. Ji*, W. Liang, D. L. McPherson, and K. Driggs-Campbell. "Structural Attention-based Recurrent Variational Autoencoder for Highway Vehicle Anomaly Detection," IFAAMAS International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2023
[Paper] [Website] [Code] **(Accepted as full paper with 23.3% acceptance rate and received student scholarship)**
- S. Liu, P. Chang, H. Chen, **N. Chakraborty**, and K. Driggs-Campbell. "Learning to Navigate Intersections with Unsupervised Driver Trait Inference," IEEE International Conference on Robotics and Automation (ICRA) 2022
[Paper] [Website] [Code]
- S. Liu*, P. Chang*, W. Liang+, **N. Chakraborty+**, and K. Driggs-Campbell. "Decentralized Structural-RNN for Robot Crowd Navigation with Deep Reinforcement Learning," IEEE International Conference on Robotics and Automation (ICRA) 2021 [Paper] [Website] [Code]

Workshop Papers:

- A. Hasan, **N. Chakraborty**, C. Wu, and K. Driggs-Campbell. "Towards Co-operative Congestion Mitigation," Shared Autonomy in Physical Human-Robot Interaction: Adaptability and Trust, ICRA 2022 Workshop [Paper]

Theses:

- "Detecting the Unpredictable: Advanced Trajectory and Camera-Based Anomaly Detection Methods for Vehicles," Illinois Digital Environment for Access to Learning and Scholarship (IDEALS), 2023 [Paper]
- "Hierarchical Self-Imitation Learning in Single-Agent Sparse Reward Environments," Illinois Digital Environment for Access to Learning and Scholarship (IDEALS), 2021 [Paper]

Patents Applied:

- K. Balakrishnan, **N. Chakraborty**, and D. Upadhyay. "Model-based Reinforcement Learning," US20240320505A1 [Paper]

Industry Experience:

- **PhD Research Intern in Sight Lab at Dolby Laboratories** Summer 2025
Researching greenfield project within the space of creative 3D asset generation [Website]
- **Research and Advanced Engineering Intern in Core AI/ML at Ford Motor Company** Summer 2022
Designing sample efficient model-free + model-based RL methods [Website]
Outperforming classical PID controllers by 41% in complex autonomous vehicle use case
Conducting both independent and collaborative research and guiding experimental design according to results
- **Perception Engineering Intern in Autonomy Team at Brunswick i-Jet Lab** Summer 2021
Localizing swimmers around boats using filtering, tracking, and computer vision techniques [Website]
Researching sensors and communicating with sensor companies to increase autonomy stack capabilities
Analyzing functional safety standards practiced at company and presenting findings to global management team
- **Engineering Intern in Global CAD Team at Qualcomm** Summer 2020
Building generalized data gathering solutions to support Design For Test pipeline [Website]
Leading design process of a base framework for data gathering tool
Collaborating and adapting with international teams to consider multiple perspectives

- Global Management Trainee Intern in Solutions at Anheuser Busch** Summer 2019
 Identifying root causes of a multi-million-dollar annual problem with Six Sigma LEAN exercises [\[Website\]](#)
 Implementing short- and long-term process solutions leveraging technology with an annual ROI of \$1.5M
 Pitching solutions to multidisciplinary teams in the People department including managers and directors
- Global Management Trainee Intern in Logistics at Anheuser Busch** Summer 2018
 Increasing productivity of critical decision-making team by developing clear visualizations [\[Website\]](#)
 Creating effective data visualizations with Qlik Sense and SQL databases
 Connecting with multidisciplinary logistics teams and interns

Teaching Experience:

- Lab Teaching Assistant for Introduction to Robotics (ECE 470)** Fall 2021 – Fall 2022
 Guiding students to program a UR3 arm with ROS and implement kinematics and computer vision
- Undergraduate Course Assistant for Digital Systems Laboratory (ECE 385)** Fall 2019 – Spring 2021
 Providing impactful assistance to students on TTL and System Verilog hardware labs

Coursework:

- Artificial Intelligence**
 - Intro to Artificial Intelligence** search, classification, NLP, CV, robotics
 - Intro to Deep Learning** linear classifiers, MLPs, CNNs, RNNs, generative models, deep RL
 - Intro to Reinforcement Learning** RL foundations, model-free, policy gradient, exploration/exploitation
 - Pattern Recognition** nearest neighbor, regression, optimization primal/dual, SVM, learning theory
 - Generative AI Models** normalizing flows, VAEs, GANs, RNNs, transformers, applications, explainability
 - Random Processes** measure theory, convergence of sequences, estimators, random walks, martingales
 - Computer Vision** image processing, fitting, geometric vision, recognition
 - Statistical Learning Theory** concentration inequalities, empirical risk minimization, generalization bounds
- Robotics**
 - Intro to Robotics** rigid motion, forward/inverse kinematics, motion planning, vision
 - Human-Centered Robotics** design robots that interact with people safely
 - Principles of Safe Autonomy** path planning, localization, lane detection, safety verification
 - Control Systems** dynamic models and response, root locus/frequency response techniques
 - Control System Theory & Design** state space models, stability, controllability, observability, tracking
 - Hybrid Systems and Control** stability, observability, etc. of switched and hybrid control systems
- Hardware/Software Systems**
 - Digital Systems Lab** logic types, storage, design tradeoffs, FPGAs, microprocessor design
 - Computer Systems Engineering** operating system design, I/O, synchronization, interrupts, virtualization

Additional Projects:

- Efficient FPGA Smart Home Security Camera System (Project Watchdog)** Apr 2019 – Oct 2019
 Leading hardware design of accelerated IoT security system
 Student Group Project
 Regional Finalist in InnovateFPGA 2019 [\[Website\]](#)
- Brain Computer Interface Platform for IoT Applications (Project HackMe)** Feb 2019
 Leading data analytics and storage team
 Student Group Project
 HackIllinois 2019 runners-up and sponsor award winners [\[Website\]](#)
- Human interactive Balancing Security Robot** Aug 2016 – Jun 2017
 Advisor: Dr. George Kantor
 Carnegie Mellon University
 Engineering a human interactive segway security robot to roam halls of an institution for safety [\[Website\]](#)

- Robotics Project at ZeGoBeast LLC Pittsburgh**

May 2016 – Jun 2017

Advisors: Mr. Daniel Goncharov and Mr. Alex Thomson
 Building & improving the wooden ZeGoBeast Electric robot and presenting at New York Maker Faire

ZeGoBeast LLC Pittsburgh

[\[Website\]](#)

Organizations & Extracurriculars:

- Eta Kappa Nu (HKN) Electrical Engineering Honor Society**

Aug 2019 –

Dedicated to serving the ECE & Engineering student body by providing services to help students succeed

Member
- Human-Centered Autonomy Lab**

Aug 2021 – Aug 2024

Connecting passionate undergraduate researchers with graduate mentors

Research Volunteer Coordinator
- iRobotics Combotics & Projects Student Organization**

Aug 2017 – May 2018

Collaborated on the mechanical design team by developing CAD designs that model the real robot

Member
- Children's Library of Pittsburgh**

Jun 2012 – Jun 2017

Shelved, counted, & organized books in the Children's section of the Main Library

Volunteer
- Supported in the planning & development of tech programs to introduce children to programming

Academic Honors and Services

Awards:

- IROS Last-Mile Robotics Workshop Best Poster 2023
- AAMAS Student Scholarship 2023
- Graduated BS with Honors 2021
- Eta Kappa Nu Member 2019 – present
- AB InBev GMT Full Tuition Scholarship 2017 – 2021
- James Scholar 2017 – 2021
- Dean's List 2019 – 2021
- InnovateFPGA Regional Finalist 2019
- HackIllinois 2019
 - 2nd Place
 - Caterpillar Sponsor Award
 - Particle Sponsor Award

Students Mentored:

- Abdulrahman Almana 2024; KAUST Gifted Student Program at UIUC
- Cassidy Wall 2024; BS Systems Engineering and Design
- Yixiao Fang 2023 – 2024; BS Computer Engineering 2023; Now studying MS at UIUC
- Chun-Yen Yeh 2024; MEng Autonomy & Robotics
- ChuLin Huang 2024; MEng Autonomy & Robotics
- Louis Sungwoo Cho 2023; BS Civil and Environmental Engineering 2023; Now studying MS at UIUC
- Wenjian Zhao 2023; BS Engineering Mechanics 2023; Now studying MEng at MIT
- Devika Prasad 2022 – 2023; BS Computer Science, Now at Applied Intuition
- Haomiao Chen 2022 – 2023; BS Physics; Now studying PhD at Cornell
- Blerim Abdullai 2022 – 2023; BS Electrical and Computer Engineering 2023; Now studying MS at U of T
- Nikil Ravi 2022 – 2023; BS Computer Science 2023; Now studying MS at Stanford

Journal and Conference Reviews:

- IEEE International Conference on Robotics and Automation (ICRA) 2022, 2023, 2025, 2026
- Conference on Robot Learning 2025
- IEEE International Conference on Intelligent Robots and Systems (IROS) 2025
- IEEE MultiMedia 2025
- IEEE Robotics and Automation Letters (RAL) 2023, 2024
- IEEE Transactions on Control Systems Technology (TCST) 2024
- Springer International Journal of Social Robotics 2024
- IEEE Transactions on Network Science and Engineering (TNSE) 2023